WORKED SOLUTION

STEP I 1992, Question 16

Probability and Statistics • Network Reliability

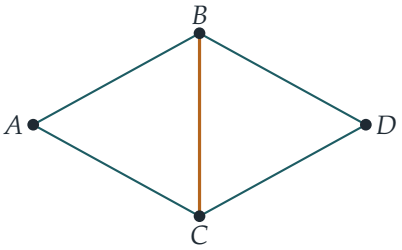
STEP I 1992, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q16

The key idea

The road network is a "bridge": two routes from A to D, one through B and one through C, with the road CB joining them in the middle. The cleanest way to handle the bridge is to condition on whether CB is open, because once we know that, everything reduces to roads in series and in parallel. Snow-proofing a road has the same effect as fusing its two ends into a single town, so we can compare the two proposals by looking at the smaller networks they create.



Solution

Throughout, write $q = 1 - p$ for the probability that a given road is open. The roads are blocked independently, so the probability of any particular pattern of open and blocked roads is a product of ps and qs. There are four routes from A to D that don't revisit a town:

$$ABD, \qquad ACD, \qquad ABCD, \qquad ACBD.$$

The only open route is $ABCD$

For $ABCD$ to be open we need AB, BC and CD all open. For it to be the *only* open route, ABD must be blocked, and since AB is open this forces BD to be blocked. Similarly ACD must be blocked, and since CD is open this forces AC to be blocked. Then $ACBD$

is automatically blocked too (it uses both AC and BD).

So the event is exactly "AB, BC, CD open and AC, BD blocked". By independence its probability is

$$q^3p^2 = p^2(1-p)^3,$$

as required.

The probability of getting through at present

Let E be the event that A and D are joined by unblocked roads, and condition on the road CB.

If CB is open, then B and C behave like a single town. We get from A to that town unless both AB and AC are blocked, and from it to D unless both BD and CD are blocked. These involve different roads, so

$$\mathrm{P}(E \mid CB \text{ open}) = (1-p^2)^2.$$

If CB is blocked, the only possible routes are ABD and ACD, each open with probability q^2, independently since they share no road. So

$$\mathrm{P}(E \mid CB \text{ blocked}) = 1-(1-q^2)^2.$$

It is easier to work with the probability that A is cut off from D. Since $1-(1-p^2)^2 = p^2(2-p^2)$ and $1-q^2 = p(2-p)$,

$$\begin{aligned}\mathrm{P}(\text{cut off}) &= q\,p^2(2-p^2) + p\cdot p^2(2-p)^2\\ &= p^2[(1-p)(2-p^2) + p(4-4p+p^2)]\\ &= p^2\,(2+2p-5p^2+2p^3).\end{aligned}$$

Which road to snow-proof

Snow-proofing CB. Now CB is always open, so we are permanently in the first case above:

$$\mathrm{P}(\text{cut off}) = 1-(1-p^2)^2 = 2p^2-p^4.$$

Snow-proofing AB. Now A and B act as a single town, and we are cut off from D exactly when BD is blocked *and* C offers no way through. The way through C needs CD open and at least one of AC, CB open, so it fails with probability $p+q\,p^2$. Hence

$$\mathrm{P}(\text{cut off}) = p\,(p+qp^2) = p^2+p^3-p^4.$$

Ask yourself Both answers start with a p^2 term. How do we decide which is smaller for every p?

Subtract them:

$$(2p^2-p^4)-(p^2+p^3-p^4) = p^2-p^3 = p^2(1-p) > 0 \quad \text{for } 0<p<1.$$

So snow-proofing AB always leaves the smaller chance of being cut off, whatever the value of p. The government should choose AB; the two choices are never equally good.

This matches intuition. Fixing CB only helps when it is needed as part of a detour, but fixing AB makes A and B a single town, which is joined to D by the direct road BD and also by the path through C.

The final part

With $p = \frac{1}{10}$, the present probability of being cut off is

$$\frac{1}{100}\left(2 + \frac{2}{10} - \frac{5}{100} + \frac{2}{1000}\right) = \frac{1}{100} \times 2.152 = 0.02152.$$

This is greater than $\frac{1}{100}$, so the government will go ahead (and, by the previous part, it will snow-proof AB).

The main term $2p^2 = 0.02$ already settles it: there are two ways of being cut off by just two blocked roads, namely $\{AB, AC\}$ and $\{BD, CD\}$.

Checks

- As $p \to 0$, the cut-off probability is about $2p^2$, which counts the two pairs of roads whose blocking isolates A or D. After snow-proofing AB only the pair $\{BD, CD\}$ remains, giving about p^2, as our formula says. After snow-proofing CB both pairs remain, giving about $2p^2$.
- At $p = 1$ the present formula gives $1 \cdot (2 + 2 - 5 + 2) = 1$: if every road is blocked we are certainly cut off.
- An exact enumeration over all 2^5 patterns of open and blocked roads confirms every formula above for several values of p. At $p = \frac{1}{10}$ it gives 0.02152 now, 0.0109 after snow-proofing AB and 0.0199 after snow-proofing CB. A simulation of 2×10^5 winters gave 0.0215.

Common slips

- Adding the probabilities of the four routes being open. The routes share roads, so these events overlap and the sum overcounts.
- In the first part, forgetting that "the only open route" forces AC and BD to be blocked, and giving just $(1 - p)^3$.
- Comparing the two proposals at $p = \frac{1}{10}$ only. The difference $p^2(1 - p)$ shows that AB is better for every p.

Beyond STEP

This is a small example from *network reliability*, and the trick of conditioning on one edge (or fusing its ends) is the deletion–contraction idea that runs through graph theory. For large random networks the same questions lead to *percolation theory*, where there is a sharp threshold value of p at which long-range connections suddenly disappear. Grimmett and Welsh, *Probability: An Introduction*, is a good place to build the conditioning tools used here.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.